

Mimicking some features of a root-soil system by dual porosity media

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ARTICLE INFO

Keywords:

Evaporation
Porous media
Dual porosity
Root-soil coupling

ABSTRACT

We report an experimental study of evaporation from a non-homogeneous dual porous system consisting of a finer pore size material at the centre surrounded by coarser particles in a glass beaker. This configuration can be considered to mimic, to a first order, the coupling between the plant roots and soil water. Similar to how roots extract water from the soils, here also water moves, driven by the capillary pressure gradient, from larger to the smaller voids. We explore two different configurations (a) when the small pore sizes span the entire porous column along the gravitational direction and (b) when they are limited to a thin annular region at the porous medium top surface. We measured the mass loss from the system, surface temperature using a thermal camera, and the ambient conditions. We compared the results with two standard systems – (a) a homogeneous porous medium and (b) a texturally layered configuration of finer particles at the top and coarser at the bottom. Compared to the homogeneous case, the duration of stage 1 (high evaporation rate regime) of evaporation and thus the characteristic evaporative length increased significantly in the case of root-mimics; this outcome shows the ability of the root-mimics in sustaining the surface water content induced by the strong capillary pressure gradient in such a dual porosity system. Finally, we track the evaporation sites using the fluorescein dye particles method and assess its effect (if any) on the overall evaporation characteristics.

1. Introduction

The relation between plant roots and soil is important not only from the perspective of agriculture but also as a fundamental understanding of the coupling between the two. On the hindsight, this coupling is crucial as it governs evapotranspiration, which essentially is an inevitable part of the global water cycle.

In field, the movement of water is very complex. Imagine a portion of soil, which is fully saturated (all the pores inside are filled) with water. There are three ways in which water can move out of this region. One, gravitational forces may drain some water depending on the soil type (Veihmeyer and Hendrickson, 1931; Israelsen and West, 1922). It is believed that within two to three days, a soil achieves field capacity (Richards and Weaver, 1944), which is defined as the percentage of water present in the (usable) soil after gravitational drainage. Two, evaporation leads to continuous loss of moisture from the soils. Such an evaporation phenomenon depends on many factors like particle sizes (Lehmann et al., 2008; Yiotis et al., 2012; Assouline et al., 2014; Kumar and Arakeri, 2018a), particle (or pore) size distribution (Lehmann et al., 2008; Assouline et al., 2014), particle layering (Assouline et al., 2014; Huang et al., 2013; Kumar and Arakeri, 2018b; Shokri et al., 2010), incident heat load, ambient conditions (Ben Neriah et al., 2014), and

porosity (Kumar and Arakeri, 2018a). Essentially, soil evaporation is determined by preferential air invasion and subsequent water transport at all times (Or et al., 2013). Three, roots or root hairs, tiny in size, suck out water from the soils, thus reducing moisture content in the soils. All the three processes together govern the overall dynamics of moisture loss from soils. Apart from these three ways of water transport, other indirect factors, like rainfall and erosion, are also important. Note that apart from acting as a source of water, soils also act as a support system for the plants.

The coupling between roots and soil moisture is crucial in understanding the movement of water to and from soils. We mimic this coupling using a dual porosity system; a lower particle size material at the centre surrounded by larger particles. The lower particles mimic the roots while the larger represent soils. In this study, gravitational drainage has been avoided but its inclusion is straightforward. The other important feature of the present study is that the finer particle size material is kept flush with the surrounded porous medium viz. both the surfaces are at nearly the same horizontal level. This study is, therefore, a first order understanding of plant root and soil water coupling. The real root-soil coupling is highly complicated, which are extremely difficult to mimic in the laboratory experiments; take the complex structural patterns and geometry of the root hairs spreading out of the main

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<https://doi.org/10.1016/j.agwat.2019.105855>

Received 8 June 2019; Received in revised form 7 October 2019; Accepted 8 October 2019

Available online 04 November 2019

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root as an example. In actual situations, roots supply water to various parts of a plant (or a tree) such as leaves, where mass loss (transpiration) happens through small openings called stomata. In the present investigation, we do not intend to mimic the entire plant system though stomatal transpiration in its own right is interesting (Kumar and Arakeri, 2019a; Kato et al., 2004).

2. Materials and methods

The experiments, reported in this study, were carried out in a glass beaker (90 mm high and 63 mm inner diameter). We mimic the plant roots with two different materials (i) a long chalk (80 mm long with an average diameter of 1 cm) piece and (ii) a thin porous medium (0.20–0.30 mm diameter glass beads). The surrounding soil is a porous medium consisting of 0.70–0.80 mm diameter glass beads. The root-mimics were placed at the centre of the surrounding porous medium as

seen schematically in Fig. 1a and b. An infrared heater, distanced ~ 19 cm away from the porous medium top surface, was used to boost the rates of evaporation. The mass lost from the system was continuously monitored using a precision weighing scale (Sartorius GPA5202), connected to a computer. The ambient conditions (temperature using a T-type thermocouple and relative humidity using HIH-3000 Honeywell sensor) were regularly (15 s interval) monitored. A thermal camera (Fluke Ti400, 320×240 pixels) placed above the medium measured its top surface temperature. The schematic of the experimental set up is seen in Fig. 1e. The container was insulated at the impervious sides to reduce heat leak from the system. The incident heat load on the porous media top surface was $\sim 1000 \text{ W/m}^2$ in all the cases, this value is obtained after extrapolation of data in a previous study (Kumar and Arakeri, 2015).

We now give a description of the experiments, whose details are seen in Table 1. As mentioned earlier, we used two different

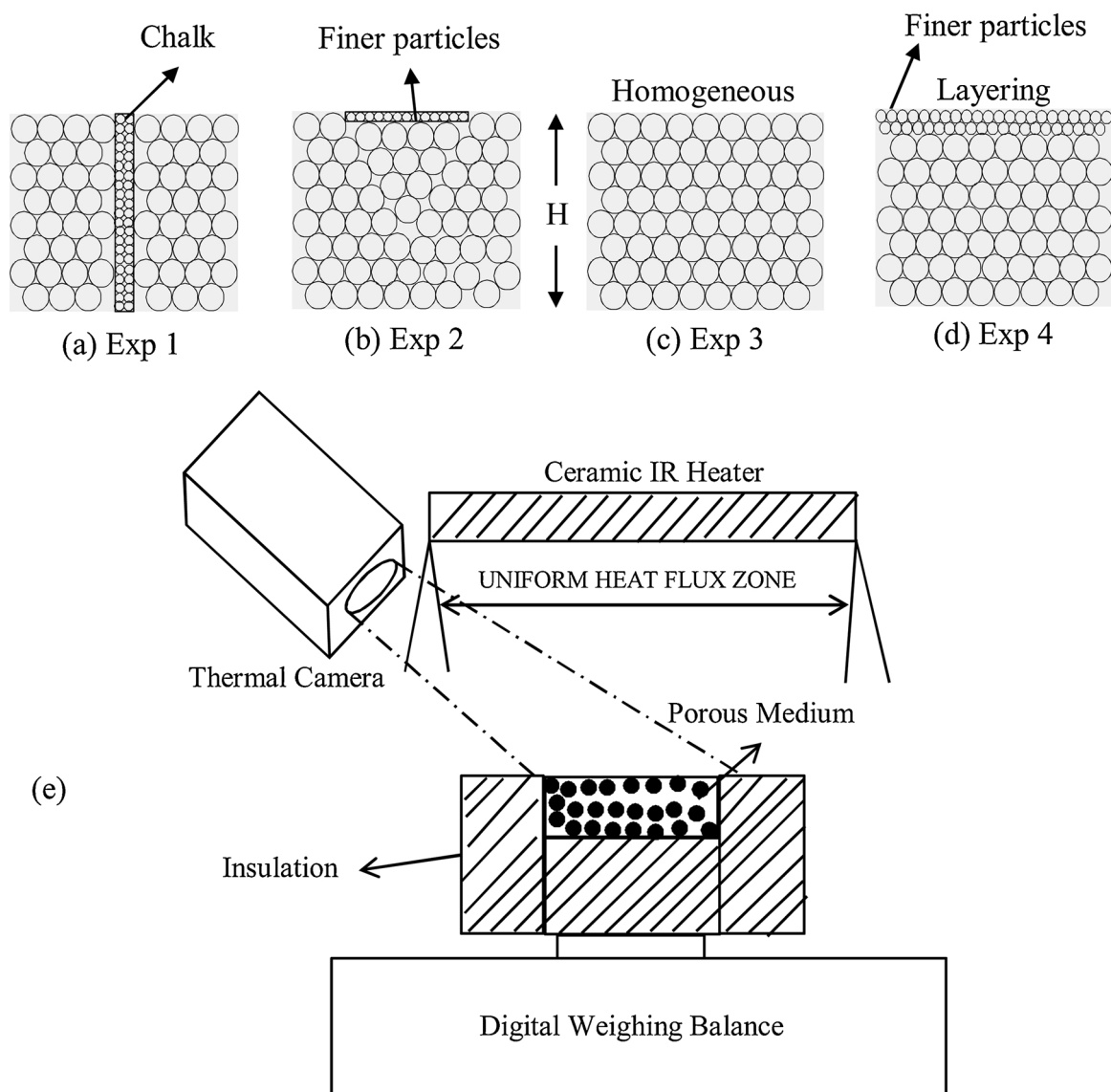


Fig. 1. The four different types of experiments (a–d) conducted in the present study. Configurations (a) and (b) can be considered as a simple approximation of the behaviour of plant roots. ‘H’ denotes the porous column height and not the container height. A schematic (e) of the experimental set up to study the coupling between root mimics and the soil mimics.

Table 1

Materials and their dimensions used in the present study. Note the values of the overall porosity, which is nearly same in all the cases.

Exp	Centre piece dimensions	Surrounding piece dimensions	Overall porosity
'1' – Dual porosity	Material : chalk Diameter : 1 cm Surface extent : 1 cm ϕ Depth : 80 mm	Material : glass beads Diameter : 0.70–0.80 mm Column Height : 80 mm	34.86%
'2' – Dual porosity	Material : glass beads Diameter : 0.20–0.30 mm Surface extent : ~ 35 mm ϕ Depth : ~ 3 mm	Material : glass beads Diameter : 0.70–0.80 mm Column Height : 80 mm	34.98%
'3' – Homogeneous	Material : glass beads Diameter : 0.70–0.80 mm Surface extent : 63 mm ϕ Column Height : ~ 87 mm		35.20%
'4' – Layering	Configuration : Finer particles on top of coarser Finer particles diameter : 0.20–0.30 mm Finer particles depth : ~ 10 mm Coarser particles diameter : 0.70–0.80 mm Coarser particles depth : ~ 73 mm Total column Height : 83 mm		34.88%

configurations to mimic the root-soil water coupling; height of the porous column (H) was ~ 80 mm in both the cases. In the first case, we used a chalk ('Exp 1'), whose pore sizes are several orders of magnitude lower than the surrounding 0.70–0.80 mm diameter glass beads. Here the chalk was placed at the centre and the larger beads were slowly packed with water in steps (Kumar, 2017). The chalk was in contact with the larger beads throughout its height (Fig. 1a). In the second case, the height of the finer particles ('Exp 2') used was negligible compared to the porous column height (Fig. 1b). The porous media considered here is a 3-D system due to a large number of layers in any direction; this also ensures limited boundary effects. In both the cases, the finer particles are expected to behave as roots since they would suck water from the larger particles. We compare the results of the present study with a reference system (Exp '3') consisting only of the larger (0.70–0.80 mm diameter glass beads) particles. Note that a few studies (Shahraeeni and Or, 2010; Aminzadeh and Or, 2014) exist where mass loss from a system, as the one in Exp '1', was investigated. A similar system could be finer particles separated by coarser either vertically or horizontally like in 'Exp 4' (see Table 1). These systems (Kumar and Arakeri, 2018b; Lehmann and Or, 2009) too offer preferential movement of water but they cannot be considered as simulating root soil water coupling due to their large surface extents. We believe that Exp '2' is a relatively new concept, which may be closest to simulating the (grass-type) root-soil water coupling, given the negligible depth of the finer particles compared to the porous column height. Note that 'Exp 2' is different than 'Exp 4' in two main aspects (a) surface extent of the finer particle sizes is lesser in 'Exp 2' and (b) the depth of finer particles in 'Exp 2' is much lower compared to that in 'Exp 4'.

2.1. Geometrical and particle-scale features of the chalk

Chalks come in various sizes and shapes, cylindrical being the most common. The chalk used in the present study is seen, schematically, in Fig. 2a. It is nearly cylindrical with small (0.6° from the axis) amount of tapering along the length. The chalk diameter at one end was 8.42 ± 0.08 mm, at the other end was 10.06 ± 0.06 mm, and its length was 79.80 ± 0.12 mm; the standard deviation was obtained after the measurements were taken for six chalk samples. The dry density of the chalk was 1.14 ± 0.02 g/cc. The overall porosity of the chalk sample was obtained in a separate experiment. The chalk piece was dipped

(slowly) in DI water, kept in a container, along its length. The water level (~ 6 mm) in the container was kept such that the chalk piece did not submerge completely; this was done to avoid any air entrapment during this (imbibition) process. After ~ 2 min, the completely wet chalk was carefully removed such that any extra water drained-off; during this duration, the chalk did not lose any significant mass due to dissolution of its particles in water. The difference in the mass readings of the wet and dry chalk gave us its porosity (ratio of the volumes of water and the chalk), which we calculated to be $52.53 \pm 0.42\%$.

The particle size has a broad range (Majumdar and William, 2009; Zhang et al., 2015) in a piece of chalk and can vary between $0.20\mu\text{m}$ to $10\mu\text{m}$ (Zhang et al., 2015). Due to the large variation in the particle sizes, the pore size would also have a broad range. The SEM image (Fig. 2b) shows very small particles as well as large plate-like particles; the scale at the bottom of this image indicates $10\mu\text{m}$. The image was taken at the central cross-sectional surface of the chalk and was captured at 5000x magnification. We see here that some particles are $\sim 10\mu\text{m}$ in size. At 20000x magnification, we were able to identify particle sizes as small as $\sim 300\text{nm}$; the resolution of this SEM image was low and hence is not shown here. These observations, related to the range of particle sizes in a chalk, are in line with previous studies (Zhang et al., 2015).

3. Results

First, we mention the initial conditions of the four experiments. After preparing the saturated mixture, some extra (~ 9 g) water was added in all the cases; this method gives evaporation rate in case of bare water (Kumar and Arakeri, 2018a,b,c,d; Kumar and Arakeri, 2019a). As mentioned previously, we measured the mass loss, using a weighing scale, which we converted (Kumar, 2017) to get the instantaneous rate of evaporation (E). Since we began with some extra water on the top, the porous media achieve saturation ($S = 1$) upon evaporation after some time; the evaporation rate at this instant is E_{max} . In the case of 'Exp 1', 'Exp 2', 'Exp 3', and 'Exp 4', the value of E_{max} was ~ 21.0 , 20.8 , 19.4 , and 19.8 mm/day, respectively; slight variation in E_{max} values are due to different ambient conditions (air temperature and relative humidity in the ambient air) in these experiments. Note that when $0 < S(t) < 1$, we get $E < E_{\text{max}}$. We use this to understand (see Fig. 3) variations of the dimensionless evaporation rate (E/E_{max}) with the depth of the average water level at any instant, Z calculated as the product of $(1 - S)$ and H i.e. $Z = H[1 - S(t)]$. The value of $Z (=Z_0)$ at which stage 1 ends has been defined (Kumar and Arakeri, 2018a, b; Kumar, 2017) as an evaporative capillary characteristic length (L_{cap}) i.e. $L_{\text{cap}} = Z_0 = H(1 - S_0)$, where S_0 is the value of saturation at the end of stage 1 of evaporation. Note that saturation values are calculated as the ratio of the values of the remaining mass of water instantaneously and at $S = 1$. Fig. 3 also shows, the well-known, three stages of evaporation in all the cases; the arrows point to the end of stage 1, which was marked using sudden change in $1/E^2 - t$ curves, whose variation is shown for 'Exp 1' and 'Exp 2' in the inset.

Following major conclusions, drawn from Fig. 3, are

- Stage 1 is not a constant rate period (CRP) unlike the isothermal cases (Lehmann et al., 2008; Yiotis et al., 2012; Assouline et al., 2014; Shokri et al., 2010; Ben Neria et al., 2014; Or et al., 2013; Lehmann and Or, 2009); similar deviating trends were found previously (Kumar 2018a; Kumar and Arakeri, 2019a; Kumar, 2017; Kumar and Arakeri, 2018c). The gradually decreasing rates of evaporation in stage 1 in non-isothermal cases were shown (Kumar 2018a) to be due to the shrinking wet patches of water on the porous medium surface.
- In stage 1, the values of E/E_{max} in 'Exp 2' are slightly higher compared to the base case ('Exp 3') at any given Z . However, surprisingly, evaporation behaviour, in stage 1, of 'Exp 1' is very similar to the base case.

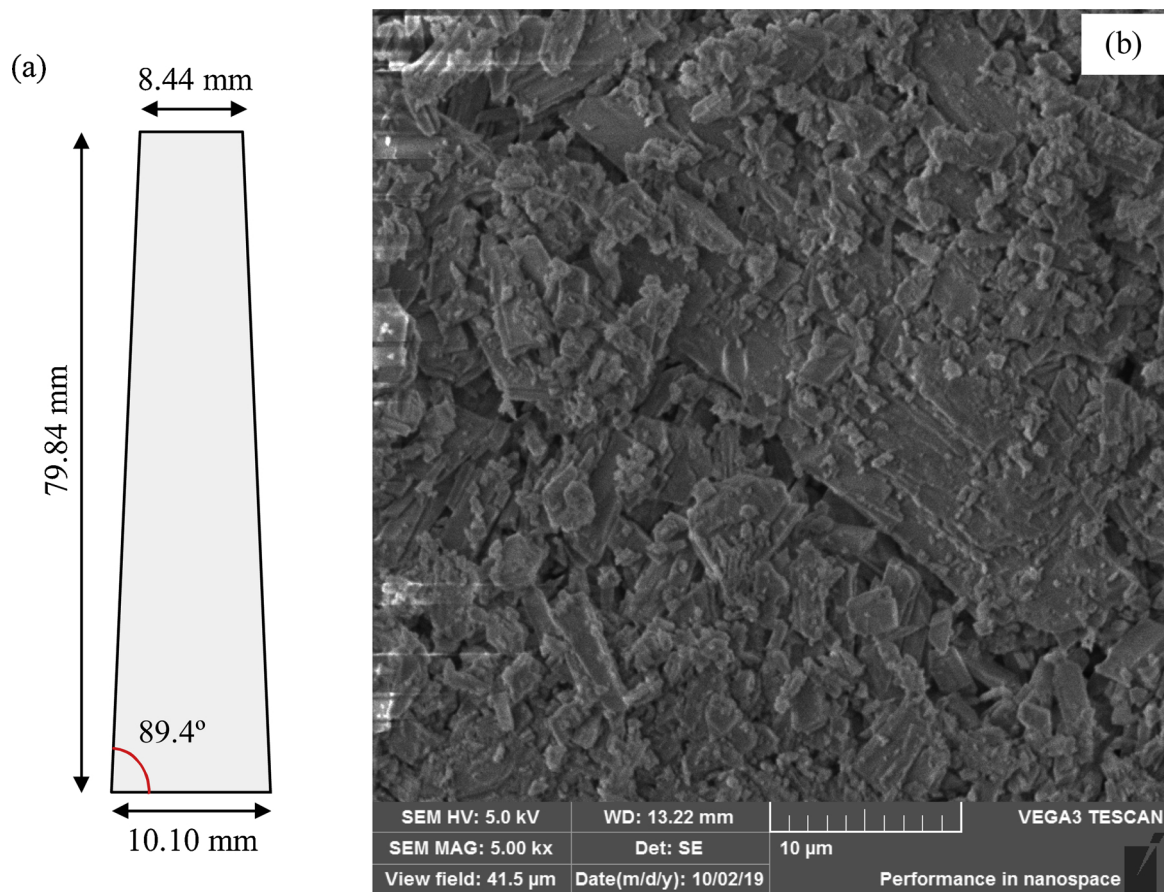


Fig. 2. Schematic (a) of a chalk piece showing its geometrical dimensions and (b) SEM image (5000x magnification) of a small ($40\mu\text{m} \times 40\mu\text{m}$) region in the central cross-sectional surface of the chalk. Range of particle sizes are seen in the micrograph.

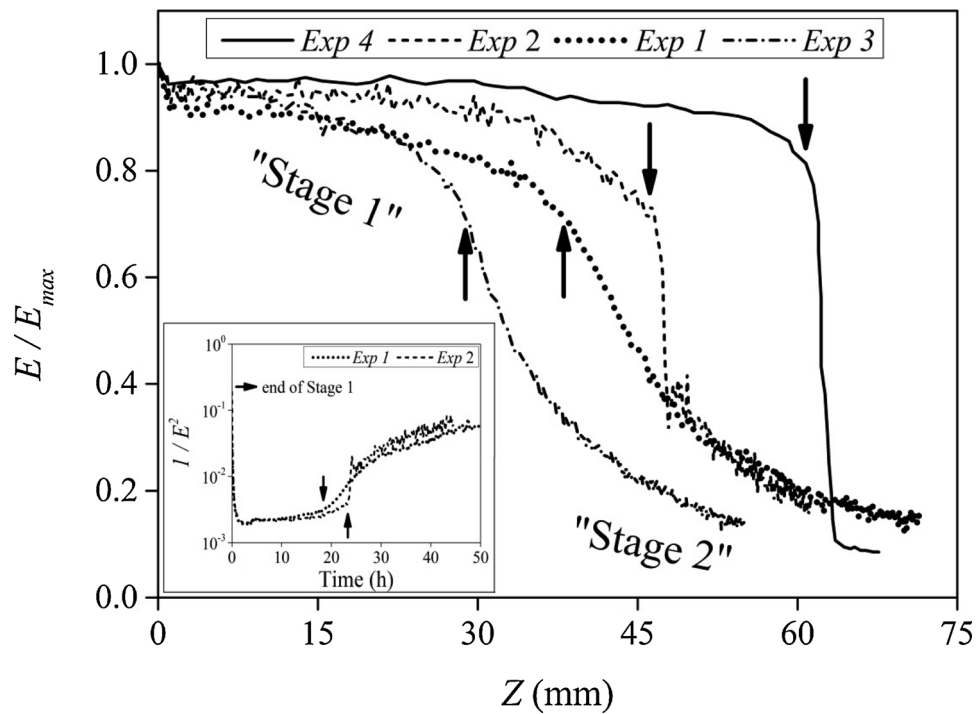


Fig. 3. Variations of the dimensionless evaporation rate with the average depth of water level for the four experiments.

- (c) Transitions are delayed in 'Exp 1' and 'Exp 2' compared to the base case.
- (d) Sudden transition in 'Exp 2' unlike being gradual in 'Exp 1' and the base case.
- (e) Results of 'Exp 4' are similar to that of 'Exp 2', which was expected since both the systems are quite similar. However, there are two significant differences (i) slightly higher E/E_{max} values and (ii) delayed transition in 'Exp 4' compared to 'Exp 2'. These differences are due to the variation in the surface extent of the finer sizes used in these cases. Finer particles cover the entire surface in 'Exp 4', which led to an extended duration of stage 1.
- (f) Consider the data in 'Exp 1' and 'Exp 3' in stage 2 of evaporation, let us say between $Z = 45$ and $Z = 54$ mm. In this range, E/E_{max} values reduced from 0.47 to 0.27 and 0.21 to 0.13, respectively. In particular at $Z = 50$ mm, E/E_{max} values are 0.32 and 0.17, respectively. The lower value of E/E_{max} in the homogeneous case is due to the fact that the vaporization plane (liquid-vapour interface) is far from the top exposed surface (surface water content is zero); the dry layers above this interface offers high resistance to the vapours. On the contrary, in 'Exp 1', even at $Z = 50$ mm, the root-mimic (chalk) was able to pull water to the top thereby maintaining some surface water content, although the surrounding medium was completely dry. Note that in a homogeneous porous medium, in stage 2 of evaporation, the vaporization plane keeps receding away from the exposed surface with time and the process of evaporation can be modelled as diffusion (Brutsaert and Chen, 1995) or diffusion-like phenomenon (Kumar, 2017).

We now discuss the evaporation process through flow visualization. For this purpose, we adopted the fluorescein dye particles method, which have been previously (Kumar and Arakeri, 2018a, b; Kumar, 2017; Kumar and Arakeri, 2018c; Weiss et al., 2018) used too. These particles are originally red, when they are dry, but turn water green in a solution. With evaporation, water is lost continuously but the fluorescein dye particles remain. On a solid surface, movement of the water-air interface (due to drying or evaporation) would leave out the dye particles, which stick to the solid surfaces; this is a continuous phenomenon. Since the particles are orange, the region of solid particles which are orange in color signifies the evaporation sites. The region of solid surfaces which are green means dye particles are still present in the solution form. This unique colour contrast is advantageous than the commonly used brilliant blue (Lehmann et al., 2008; Shokri et al., 2010; Shokri and Or, 2013) or coffee (Keita et al., 2013).

Both the experiments 'Exp 1' and 'Exp 2' were commenced with some extra water on top (see Fig. 4e). After some time we get the condition where $S = 1$ and preferential air invasion and subsequent capillary film formation occurs. The smaller voids (whether within the surrounding porous media or the root-mimic) start pulling water from the surrounding media. Fig. 4a ('Exp 2') shows the condition when the root-mimic is completely wet (green) while the surrounding medium is completely dry (orange); at this instant $E/E_{max} \sim 0.9$, $S \sim 0.64$, and $Z \sim 29$ mm ($= 0.36 \times 80$ mm) and the system is evaporating in stage 1. In Fig. 4b ('Exp 1'), everything including the chalk piece at the centre appears to be completely dry; at this instant $Z \sim 39$ mm, $E/E_{max} \sim 0.67$ and the system is at the verge of ending (see Fig. 3) stage 1 of evaporation. The colour contrast of fluorescein dye particles is useful in marking dry and wet regions. Other examples of this benefit are seen in Fig. 4c and d. The near-bottom portions in both of these images appear green (signifying fully wet region), while the near-top region is nearly fully dry. Note that the middle regions are white, which shows the true colour of the glass beads; these regions are generally drained by the capillary films. Fig. 4f shows a very interesting situation. This image was captured at the end of the experiment but the chalk is still fully wet (green). At the right hand side of this image, we observe that some dry glass beads are sticking with the chalk; this region of the chalk was exposed to the atmosphere in 'Exp 1'. Here, we see continuous

deposition of the dye particles on the beads, which should also have occurred on the near-top regions of the chalk. Note that the voids in the chalk were quite small and the deposited dye particles (after a continuous and prolonged duration of their crystallization) can easily block them. The duration of L_{cap} hence can be affected by the presence of the dye particles.

Surface water content and the movement of the liquid-vapour menisci can also be seen using thermal images (Kumar and Arakeri, 2018a, b; Kumar and Arakeri, 2019a). Here, for simplicity, we show the images for the two root-mimic experiments when the stage 1 of evaporation is about to end. Fig. 5a and b shows the respective thermal images for 'Exp 1' and 'Exp 2' cases. In both the cases, the root-mimics are much colder ($\sim 8^\circ\text{C}$) than the surrounding porous medium; the temperature colour scale and the respective details of the experiments are seen in the legends. The lower temperatures of the root-mimics clearly indicate that they are evaporating at high rates, while the high temperatures of the surrounding media indicate their dryness. Some of the heat from the hotter dry region gets conducted laterally (Kumar and Arakeri, 2019a; Aminzadeh and Or, 2014) (also beneath the top surface and within the media) and reaches the lower temperature regions and thus helps in sustaining high evaporation rates. Another common feature in these images is the similar way in which temperature decreases from the hottest container walls to the centre of the root-mimic.

Finally, we discuss the question – till what extent (or not at all) the fluorescein dye particles would have affected the evaporation characteristics? The concept of using the tracer particles lie in their use only in sparse amount. It is known (Keita et al., 2013; Nachshon et al., 2011a, b) that a large amount of tracer particles (like brilliant blue, fluorescein dye, or salt) lead to pore blockage during evaporation from a porous media; crystallization of the solid particles in such systems is an active area of research. The blockage of pores, at least near the exposed surface, affect the surface water content significantly and hence the duration of stage 1 is short i.e. transition arrives earlier. For this purpose, a separate experiment was conducted. This experiment was similar to that of 'Exp 1' but here fluorescein dye particles were not used; the experiment was repeated twice for consistency in the measurements.

Fig. 6 shows the variations of E/E_{max} vs Z for the two cases. Following major conclusions can be drawn,

- (a) Surprisingly, no significant change in the point of transition (or the end of stage 1 of evaporation) was observed.
- (b) In the stage 1 of evaporation, E/E_{max} values are slightly higher when fluorescein dye particles were not used. In the stage 2 of evaporation, E/E_{max} values are marginally higher when fluorescein dye particles were not used.

We can now say that the point of transition in the configuration used in the case of 'Exp 1' is a unique feature of such a system and the dye particles (if used only as tracers) had no effect. Slight variations (and not significant) in the rates of evaporation in these experiments could be due to the partial blockage of pores in the experiment where the dye was used. In fact, finding the critical amount of dye, which would affect the point of transition, can be studied in future; however, this is not the theme of the present investigation.

4. Discussion

The most important conclusion is the enhancement in L_{cap} values (see Fig. 3) compared to the base case. Note that L_{cap} for 0.70–0.80 mm diameter glass beads ('Exp 3') is ~ 30 mm. Roughly, the length, L_{cap} , represents the average depth, of the water level, till which a porous medium sustains stage 1. With chalk at the centre ('Exp 1'), L_{cap} increases by ~ 10 mm and with a thin layer of finer glass beads ('Exp 2') it increases by ~ 15 mm; these increased values are significant. With finer particles completely covering the coarser ones, L_{cap} increased further to

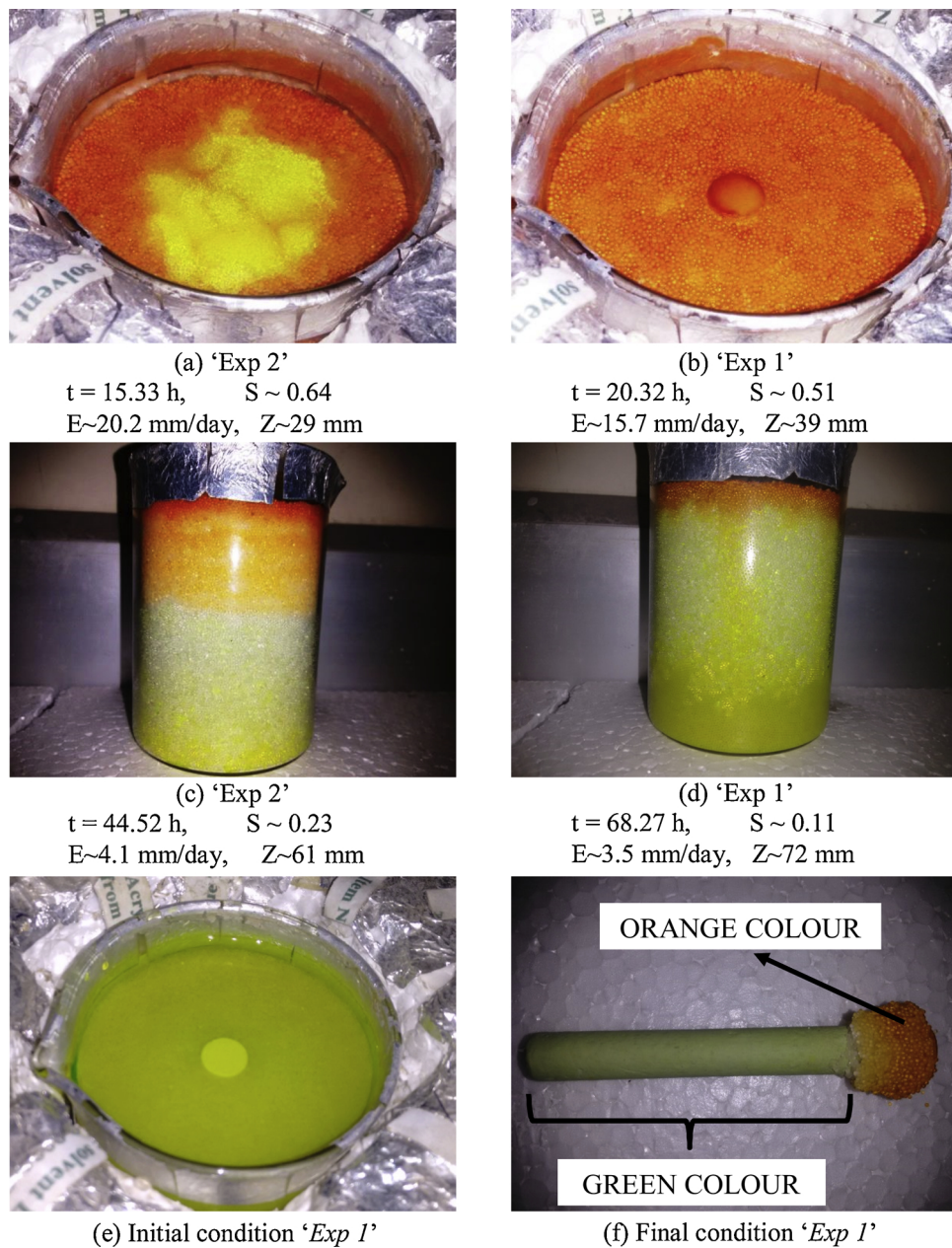


Fig. 4. Snapshots of various porous systems at different instants of the experiments. Snapshots (a), (b), and (e) were captured from the top, (c) and (d) were captured from the side, and (f) shows the chalk (green in colour) at the end of the experiment 'Exp 1' (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

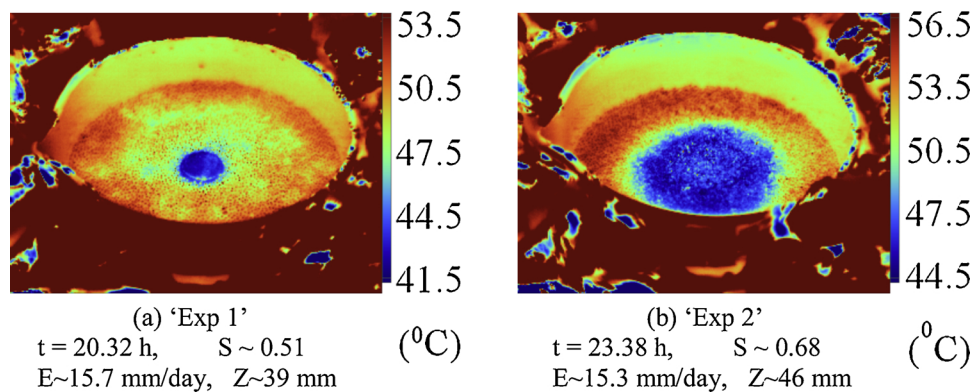


Fig. 5. Thermal images of the porous medium top surface for the two root-mimic experiments. Smaller particles at the centre successfully pull water and bring it to the top exposed surface. The water level in the surrounding larger-sized porous media is deep within it.

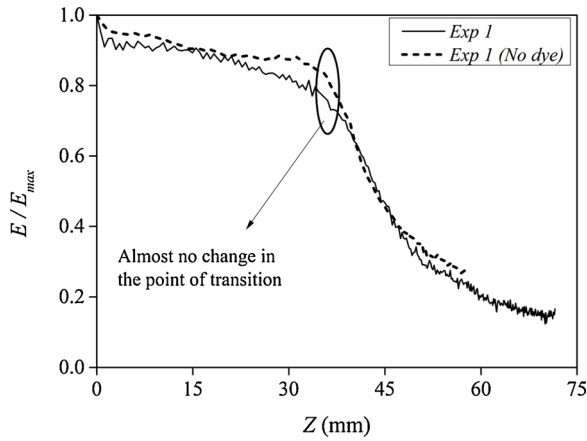


Fig. 6. Variations of the dimensionless evaporation rate with the average depth of water level for the two experiments – ‘Exp 1’ with the fluorescein dye particles and ‘Exp 1 (No dye)’ without the particles.

~60 mm. The real enhancement in L_{cap} values for ‘Exp 4’ is ~20 mm since we must remove the thickness (~10 mm, see Table 1) of the finer particles. We see that ‘Exp 1’ and ‘Exp 2’ may mimic the root-soil water coupling.

Next, we discuss the reason for the enhancement in L_{cap} values observed experimentally. Fig. 7 shows simple cartoons, for different

cases, showing liquid motion within the porous media and the liquid-vapour menisci at two different instants when the region near the top is relatively (i) wet and (ii) dry. These cartoons are only a guide and may not represent the exact scenario. Note that the initial condition in all the cases is the same viz. the system is fully saturated ($S = 1$). Consider a homogeneous system (consisting only of larger particles) as seen in Figure 7a. Just after the full saturation (let’s say $S \sim 0.95$), a dominant capillary film forms due to preferential pore air (from the atmosphere) invasion (Lehmann et al., 2008); the fluid flow is governed by the total (including capillary) pressure gradient and initially, the relatively larger pores are invaded by air. These capillary films connect the water below in the porous media to the pinned liquid on the surface. With time, these films shrink while the bulk liquid level keeps moving away from the top surface. At the critical point, the end of stage 1, the top surface loses its water content and the bulk liquid level is at L_{cap} distance away from the top surface. Films may still form within the medium but they do not reach the exposed boundary as seen in Figure 7e.

In the presence of the finer particles (whether ‘Exp 1’ or ‘Exp 2’), additional capillary films (Shokri et al., 2010; Lehmann and Or, 2009) form since the coarser particles now supply some water to the finer particle locations. Here, we have two different types of capillary films: (i) forming within the coarser particles, (ii) connecting bulk liquid to the root-mimics. At near full saturation, this scenario is seen in Figure 7b and c. Consider the instant when evaporation from the homogeneous particles is about to enter the transition regime (Figure 7e). At this

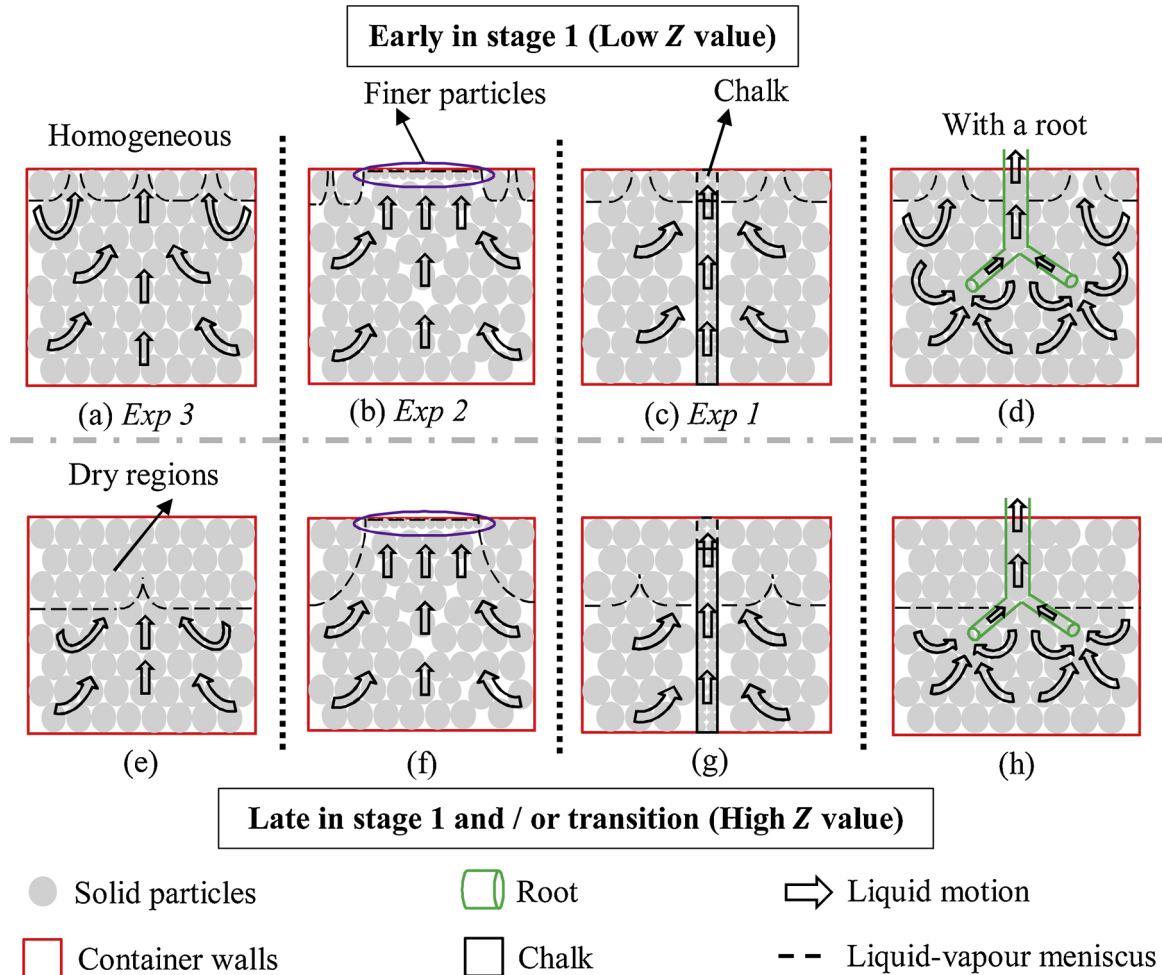


Fig. 7. Schematics showing the expected flow situation in different systems at two different stages of evaporation. Cartoons (e–g) can be considered as corresponding representatives of the conditions at, let us say, $Z = 35$ mm for the systems studied here. Cartoon (h) is a hypothetical representation of moisture transport in the presence of the real roots when the surrounding medium is already in stage 2 of evaporation i.e. it has no surface water content.

instant, films in the coarser particles do not reach the top exposed surface but the root-mimics are still able to suck water (Figure 7f and g) from the surrounding coarser particles. This additional suction extended the duration of stage 1 and L_{cap} values. In 'Exp 1' since the chalk piece spanned the entire porous column, enhancement in L_{cap} value was supposed to be larger than in 'Exp 2'. In fact, we observed the exact opposite, which we believe is due to the deposited orange fluorescein particles blocking the pores in the chalk near the surface; this is discussed soon. Even in the case of real roots, the liquid transport (Figure 7d and h) and the movement of the liquid-vapour menisci are expected to remain similar to the ones observed here.

Theoretically, the accurate prediction of enhancement in L_{cap} values are not straightforward as the liquid path is quite complicated during the entire evaporation process; one must consider possible pathways for liquid flow from largest to larger and eventually to the smaller and the smallest pores at a given instant. For example: voids formed by 0.80 mm diameter glass beads are the largest. These voids will supply water to the voids formed by all the three particle sizes (0.70, 0.30, and 0.20 mm diameter). At the same time voids between 0.70 mm diameter particles will supply water to the two smaller (0.30 and 0.20 mm diameter) voids. Finally, we must also consider water movement from voids of 0.30 mm diameter particles to voids of the smallest (0.20 mm diameter) particles. Following Lehmann et al. (2008), we calculated values of L_{cap} for the two homogeneous systems consisting of 0.70–0.80 and 0.20–0.30 mm diameter glass beads to be ~ 13 and 102 mm, respectively. When these balls are mixed in a vertical contrast, the theoretical (Lehmann and Or, 2009) value of L_c is ~ 319 mm. Using theory for the horizontal contrast (Assouline et al., 2014; Shokri et al., 2010), we calculated L_c to be ~ 16 mm. For the 'Exp 4', L_{cap} is ~ 60 mm (see Fig. 3), which lies between the theoretical values of vertical (~ 319 mm) and horizontal contrasts (~ 16 mm). With the limited surface extent and thinner depth, L_{cap} in 'Exp 2' (~ 46 mm) lies between the theoretical value in case of horizontal contrast (~ 16 mm) and L_{cap} in 'Exp 4' (~ 60 mm); this trend seems to be true intuitively also. We see that the simple models were not able to predict the enhancement in L_{cap} values correctly and the offset is quite large too. The theoretical model in a porous medium with contrast needs to be revisited for the correct adjustments. Note that, in addition to this complexity, the effect of incident heat load on mass transfer aspects of a homogeneous porous medium has recently (Kumar and Arakeri, 2019b) been studied. More recently, dual porosity and dual permeability systems have been modelled using proper orthogonal decomposition approach (Wang et al., 2017, 2018a; Wang et al., 2018b). This relatively new method to handle the compressible gas flow in a porous network has been shown to be computationally much faster and highly accurate.

5. Conclusions

Experimental investigations of evaporation from unique double porosity systems were conducted and the observations were compared with a homogeneous and a horizontally layered system. The goal was to understand the plant root-soil water coupling in the absence of gravitational drainage. The roots were mimicked using two materials (a) porous chalk with very low particle sizes spanning the entire porous column and (b) a thin near-surface layer of finer (0.20–0.30 mm diameter) glass beads. Coarser surrounding glass beads with 0.70–0.80 mm diameter were used to mimic soils. The systems were heated from above using infrared radiation to increase the rates of evaporation, which occurred only from the open top surfaces as rest of the surfaces were impervious. Three diagnostics were used simultaneously – measurement of mass, measurement of the evaporating surface temperature (captured by an IR camera), and colour-based detection of wet / dry regions (or surfaces) using the fluorescein dye particles method.

Relative to the homogeneous case, we found a substantial increase in the L_{cap} values. The increment was argued to be due to additional suction applied by the root-mimics on the larger particles when they are

hydraulically connected. We expected the chalk piece (at the centre) to have greater L_{cap} compared to the other root-mimic but it, in fact, is lesser; this experiment was also conducted (and repeated twice for consistency) where the dye particles were not used. The deposited fluorescein dye particles partially blocked the pores in chalk but had nearly no effect on the overall evaporation characteristics. The dye particles, used in small amount, acted as tracers and helped in tracking the evaporating and draining zones based on a unique colour contrast. Finally, the surface water content was indirectly indicated by the thermal images. The surface temperature and the evaporation rate of any region in a thermal image follow an inverse relation to satisfy the surface energy balance.

As far as evaporation rates in the stage 2 are concerned, the homogeneous system had the least value. For the same system, but with root-mimics (especially the deep roots like the chalk in 'Exp1') the rates of evaporation are higher; this happened because the root-mimics helped in sustaining the hydraulic connections and hence some surface water content was maintained although water level in the surrounding medium was very far from the exposed surface. This shows the ability of root-mimics in redistributing the moisture within a porous system not only in stage 1 (resulting in high L_{cap}) of evaporation but also in stage 2 (resulting in slightly higher rates of evaporation) compared to a homogeneous porous medium.

The present investigation can be further extended to the case where the surrounding medium will not be allowed to evaporate. It would be interesting to compare the L_{cap} values in these cases; they are expected to be higher than the values obtained here. Additionally, an artificial plant with leaf-type surfaces can be added to the present system to complete the plant-root-soil-water coupling. In reality, this coupling is highly complicated not only due to the extent and networking of roots but also due to gravitation drainage. However, such a simplistic approach in the present study did mimic (up to some extent) the root-soil coupling in determining the overall water distribution. A possible extension of this study could be to include a real plant root in a homogeneous or heterogeneous porous medium and observe the interactions as well as the plant root water uptake.

Declaration of Competing Interest

The authors declare no conflict of interest with any government / private organizations and funding agencies.

Acknowledgements

We thank Robert Bosch Centre for Cyber-Physical Systems for the financial assistance under the grant RBCCPS/ME/JHA/PC-0013. Funding from the Ministry of Earth Sciences, India under the grant MESO/0034 is also gratefully acknowledged. We also thank Mr Ashwin for his help during the experimentation.

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